

TABLE II
SIGNAL LOGIC LEVELS AT DIFFERENT STAGES OF RECEIVER SIDE FOR A PARTICULAR COMMAND

Command from user	Arduino output via digital pins (2, 3, 4, 5)	L293D driver inputs (2, 7, 10, 15)	L293D driver outputs (3, 6, 11, 14)
Forward	HLHL	HLHL	HLHL
Backward	LHLH	LHLH	LHLH
Right	• LLHL for 1s, then HLHL if previous command was forward • LLHL for 1s, then LHLH if previous command was backward	• LLHL for 1s, then HLHL if previous command was forward • LLHL for 1s, then LHLH if previous command was backward	• LLHL for 1s, then HLHL if previous command was forward • LLHL for 1s, then LHLH if previous command was backward
Left	• HLLL for 1s, then HLHL if previous command was forward • HLLL for 1s, then LHLH if previous command was backward	• HLLL for 1s, then HLHL if previous command was forward • HLLL for 1s, then LHLH if previous command was backward	• HLLL for 1s, then HLHL if previous command was forward • HLLL for 1s, then LHLH if previous command was backward
Stop	LLLL	LLLL	LLLL

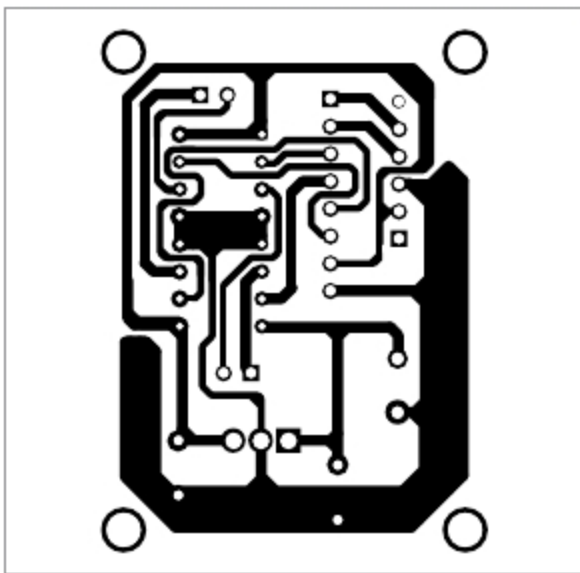


Fig. 6: Actual-size PCB layout of receiver circuit

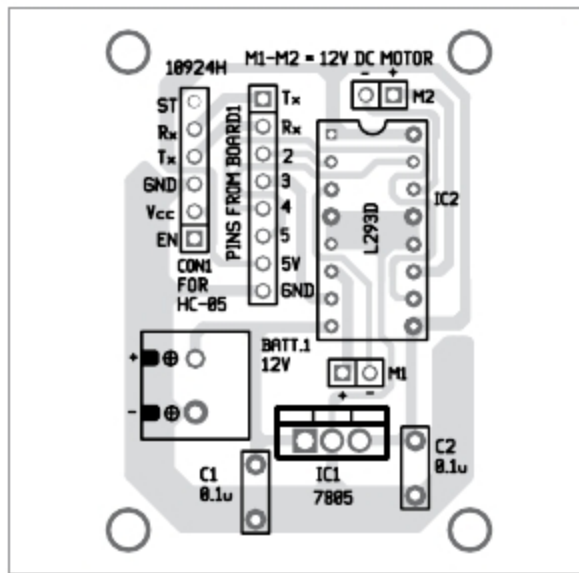


Fig. 7: Components layout for the PCB

L293D motor driver. This is a dual H-bridge high-current motor driver IC. It is used here because digital pins of Arduino cannot source enough current to drive the motors of the robotic car. H-bridges are also useful in controlling the direction of rotation of a motor. Enable pins 1 and 9 of the IC, being active high, are connected to 5V. Four output pins of L293D IC are connected to motors M1 and M2 on the receiver side.

Signal logic levels at the different stages of the circuits for proper controlling of the robotic car are given in Table II.

Arduino IDE 1.6.5 is used for programming Arduino in this project. Steps to program Arduino are as follows:

1. Select the proper COM port and

board from Tools menu in the IDE.

2. Upload source code voice_ctrl.ino to the board.

Construction and testing

An actual-size PCB layout of the receiver circuit is shown in Fig. 6 and its components layout in Fig. 7. A Micromax Q372 Android smartphone with Android Lollipop 5.0 was used to run Voicecontrol app. After successful installation of the app, turn on the Internet connection and Bluetooth of the smartphone.

Power on the receiver circuit and pair Bluetooth module with the smartphone. Default passkey is 1234. After successful pairing, open the installed app in the smartphone and press the pushbutton with Bluetooth text and logo on it. Now, the list of paired

EFY Note

The source code of this project is available for free download at source.efymag.com

PARTS LIST

Semiconductors:

- BOARD1 - Arduino Uno board
- IC1 - 7805, 5V voltage regulator
- IC2 - L293D motor driver

Capacitors:

- C1-C2 - 0.1μF ceramic disk

Miscellaneous:

- BATT.1 - 12V battery
- M1-M2 - 12V DC geared motor
- CON1 - 6-pin female Berg connector for Bluetooth module
- 8-pin female Berg connector for Board1
- 2-pin terminal connector for battery
- 2-pin male Berg connector for motors M1 and M2
- HC-05 Bluetooth module
- Android smartphone

devices will appear. Select HC-05 from the list to connect the smartphone with HC-05 Bluetooth module on the receiver side. After successful connection, 'connected' will be displayed on the main screen of Voicecontrol app. Press the pushbutton with microphone symbol and a prompt will appear asking for voice commands.

When the prompt appears, voice commands are detected by the app, which converts them into text and sends it to the receiver side wirelessly via Bluetooth. On the receiving side, Arduino checks the text. If it is a matching string, it controls the movements of the robot according to the description given in Table II. **EFY**



Saikat Patra is passionate about electronics and MCU-based embedded system applications



Shibendu Mahata is M.Tech (gold medalist) in instrumentation and electronics engineering from Jadavpur University. He has previously worked in Reliance Industries Ltd and CSIR-CMERI. He is interested in designing MCU-based real-time embedded signal processing and process control systems